

Key Architecture Components

Client-Side Interface

- **Tech Stack:** HTML5, CSS3, Vanilla JavaScript
- **Core Responsibilities:** Renders the dynamic layout, manages the interactive research workspace (including text highlighting, bookmark management, and Markdown exporting), and handles client-side visual effects like the canvas animations.

Flask API Service (`app.py`)

- **Tech Stack:** Python / Flask
- **Core Responsibilities:** Serves web templates, handles request validation, maps specific research modes to tailored system prompts, formats multi-turn chat context, and safely proxies API requests to prevent exposed credentials.

AI Integration Layer

- **Tech Stack:** Google AI Studio
- **Core Responsibilities:** Manages authentication, structures payload parameters, and dispatches generation requests to the Gemini model backends.